

Cost Analysis

2026-08-24

Contents

1	Assumptions	1
2	Executive Summary	1
3	Breakdown by grades	10
3.1	3200115	10

1 Assumptions

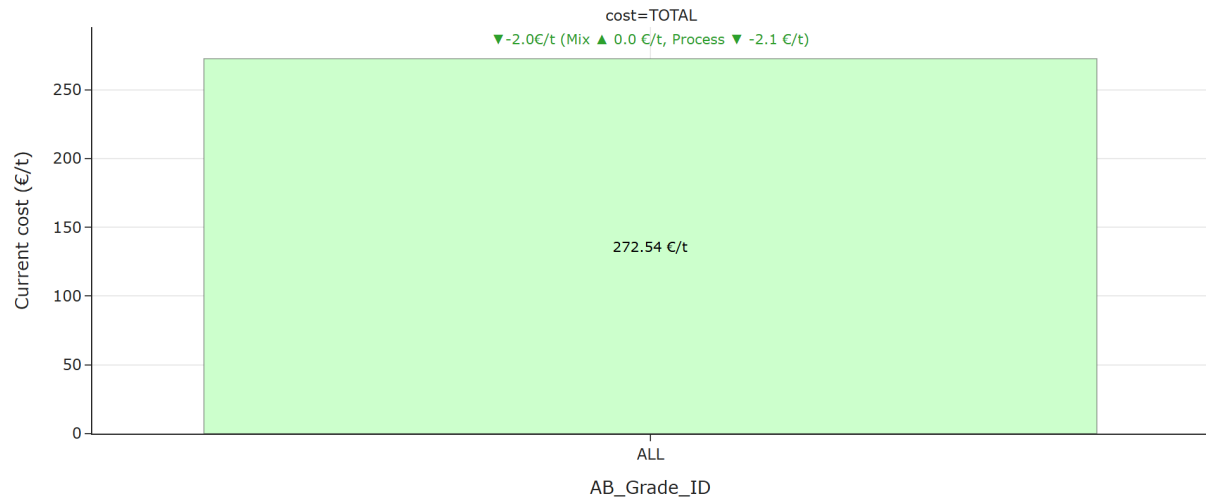
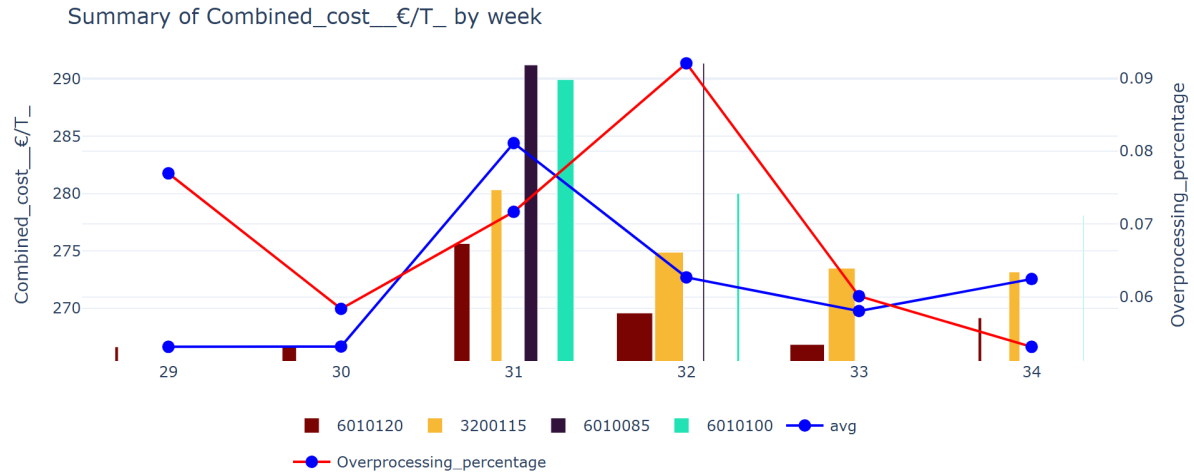
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** July 19, 2026 – August 24, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

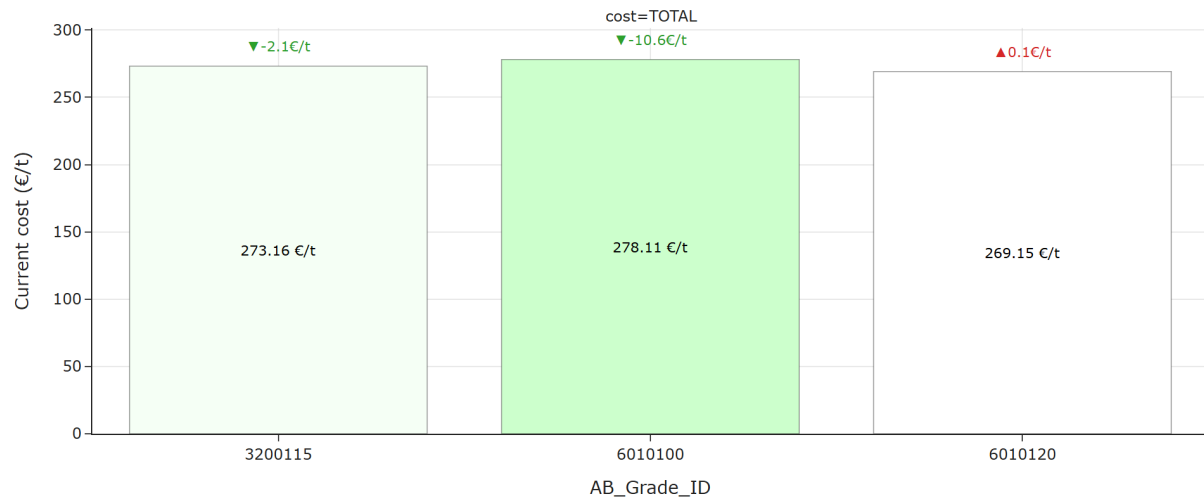
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 2.02 €/t, resulting in 272.54 €/t.

Breaking down the overall cost change, +0.04 €/t was driven by a more expensive mix of grades, and -2.06 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 2.06 €/t, resulting in 273.16 €/t.

For grade 6010100, TOTAL cost saw a decrease of 10.56 €/t, resulting in 278.11 €/t.

For grade 6010120, TOTAL cost saw an increase of 0.06 €/t, resulting in 269.15 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010120 (+0.06 €/t) and the strongest cost decrease was observed for grade 6010100 (-10.56 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam_€/T** decreased on average (-4.05 €/t).
- **Starch_€/T** decreased on average (-1.02 €/t).
- **Fibre_cost_€/T** increased on average (+0.64 €/t).
- **Electricity_€/T** increased on average (+0.14 €/t).
- **Chemicals_€/T** increased on average (+0.10 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fibre_cost_€/T** (+0.64 €/t on average), with the highest increase in grade 6010100 (+1.13 €/t) and the largest average decrease comes from **Steam_€/T** (-4.05 €/t on average), with the biggest drop in grade 6010100 (-9.93 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** decreased on average (-2.11 €/t).
- **Steam_Whitewater** decreased on average (-1.19 €/t).
- **Steam_Afterdryers** decreased on average (-0.82 €/t).
- **Steam_Heatexchangers** increased on average (+0.04 €/t).
- **Steam_Starchkitchen** increased on average (+0.03 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Heatexchangers** (+0.04 €/t on average), with the highest increase in grade 6010100 (+0.30 €/t) and the largest average decrease comes from **Steam_Predryers** (-2.11 €/t on average), with the biggest drop in grade 6010100 (-7.07 €/t). These components stand out and may require further investigation.

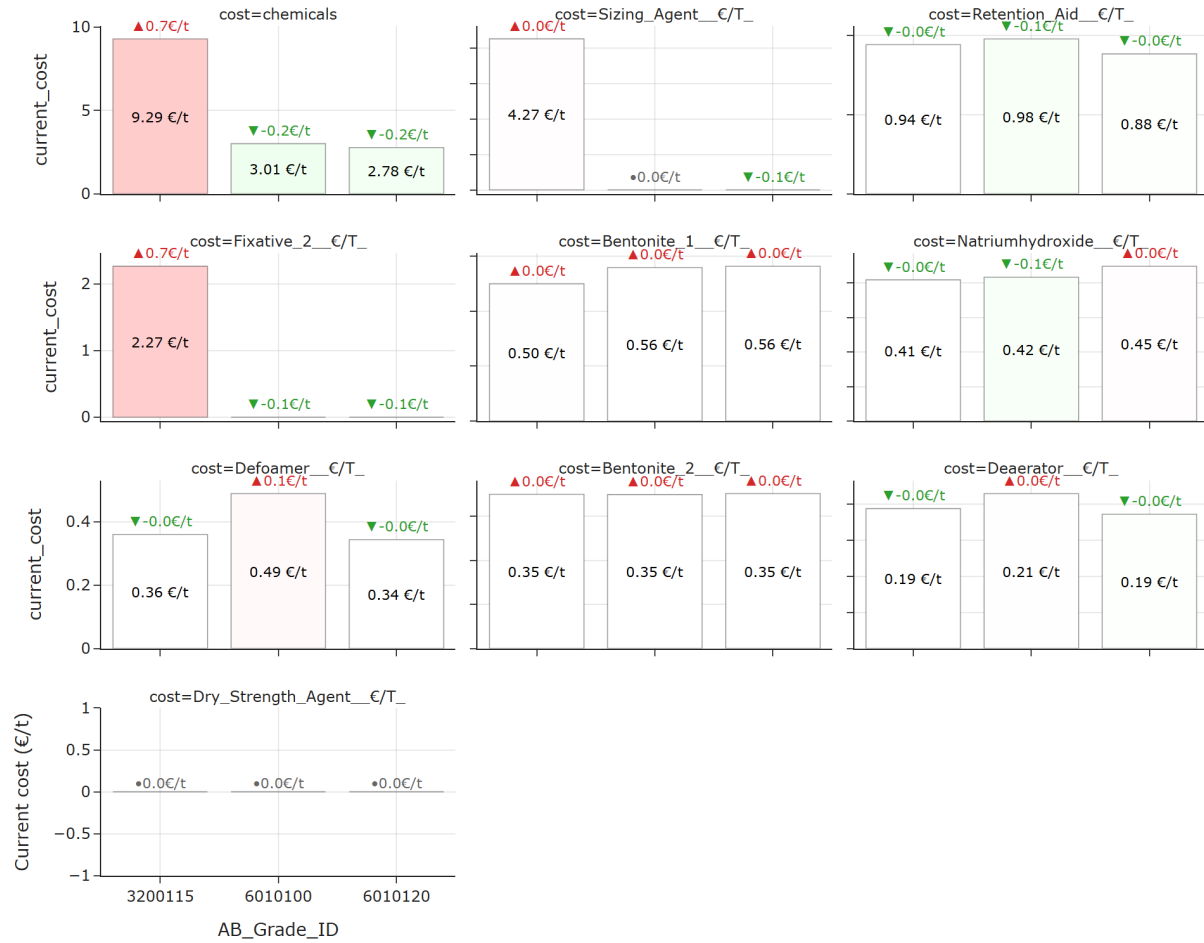


Average change by electricity:

- **Electricity_Other** increased on average (+0.11 €/t).
- **Electricity_Specific** decreased on average (-0.09 €/t).
- **Electricity_Freqconverters** increased on average (+0.08 €/t).
- **Electricity_Predryer** increased on average (+0.07 €/t).
- **Electricity_Press_and_Steambox** increased on average (+0.03 €/t).
- **Electricity_Afterdryer** decreased on average (-0.03 €/t).
- **Electricity_Wiresection** decreased on average (-0.02 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.11 €/t on average), with the highest increase in grade 6010120 (+0.18 €/t) and the largest average decrease comes from **Electricity_Specific** (-0.09 €/t on average),

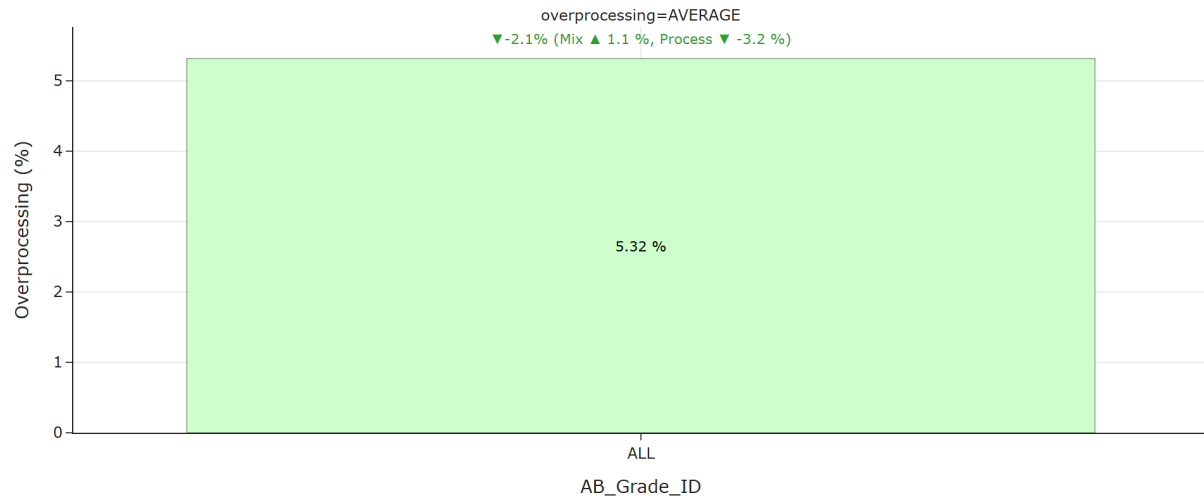
with the biggest drop in grade 6010120 (-0.63 €/t). These components stand out and may require further investigation.



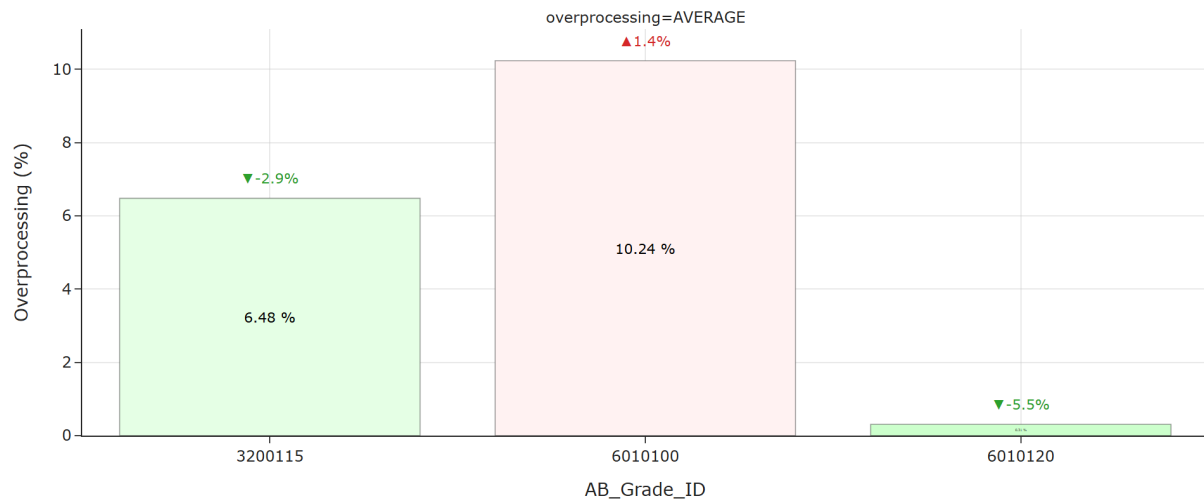
Average change by chemicals:

- **Fixative_2_€/T** increased on average (+0.72 €/t).
- **Retention_Aid_€/T** decreased on average (-0.05 €/t).
- **Sizing_Agent_€/T** increased on average (+0.03 €/t).
- **Natriumhydroxide_€/T** decreased on average (-0.03 €/t).
- **Defoamer_€/T** increased on average (+0.02 €/t).
- **Deaerator_€/T** decreased on average (-0.01 €/t).
- **Bentonite_1_€/T** increased on average (+0.01 €/t).
- **Bentonite_2_€/T** increased on average (+0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fixative_2_€/T** (+0.72 €/t on average), with the highest increase in grade 3200115 (+0.72 €/t) and the largest average decrease comes from **Retention_Aid_€/T** (-0.05 €/t on average), with the biggest drop in grade 6010100 (-0.11 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 2.11 %, resulting in 5.32 %.



For grade 3200115, TOTAL overprocessing saw a decrease of 2.87 %, resulting in 6.48 %.

For grade 6010100, TOTAL overprocessing saw an increase of 1.40 %, resulting in 10.24 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 5.53 %, resulting in 0.31 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010100 (+1.40 %) and the strongest overprocessing decrease was observed for grade 6010120 (-5.53 %). These movements stand out and may require further investigation.



Average change by component:

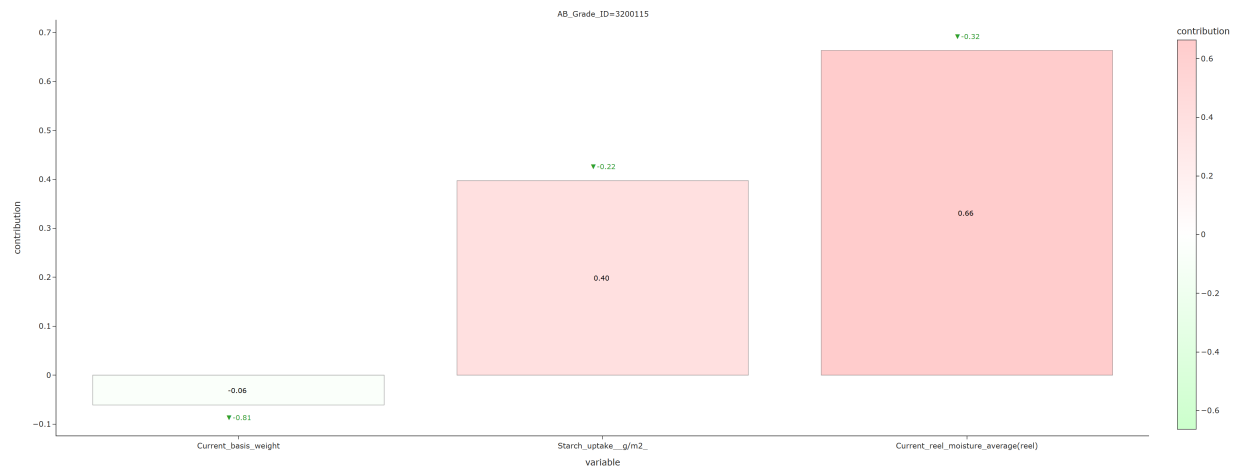
- **Burst** decreased on average (-3.79 %).
- **CMT30** decreased on average (-1.13 %).
- **SCT_CD** decreased on average (-0.27 %).

A more detailed breakdown by component shows that the largest average decrease comes from **Burst** (-3.79 % on average), with the biggest drop in grade 3200115 (-3.79 %). These components stand out and may require further investigation.

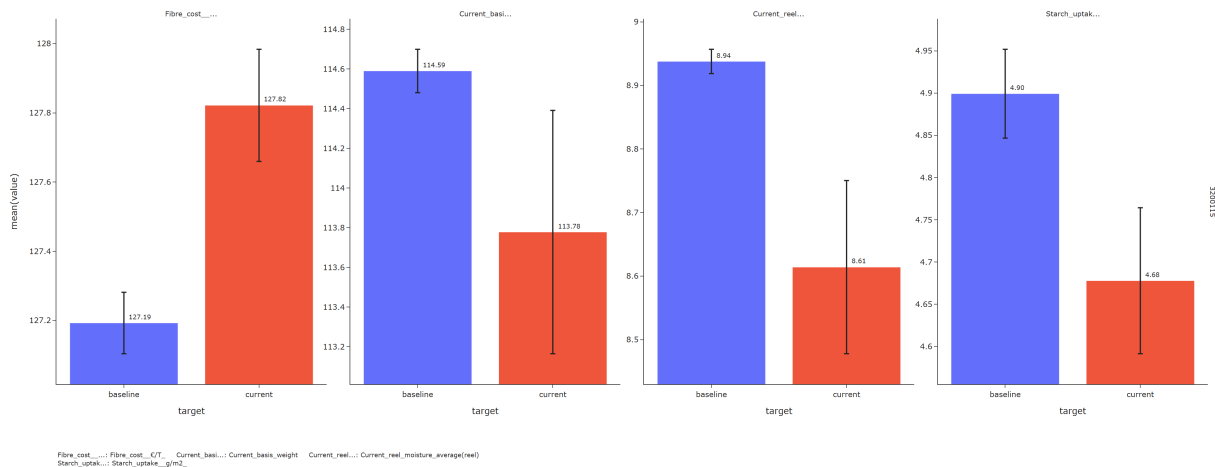
3 Breakdown by grades

3.1 3200115

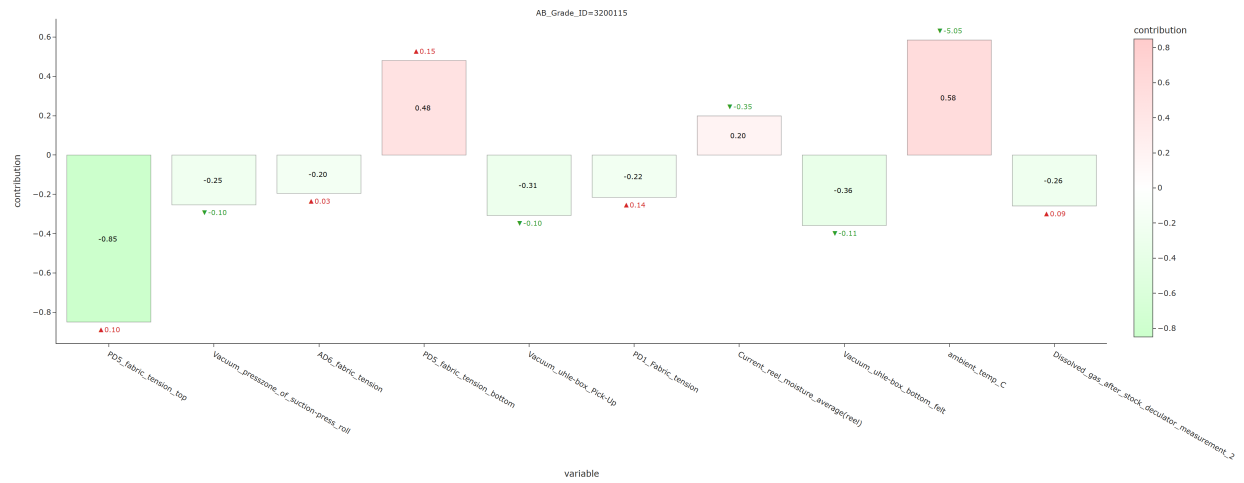
3.1.1 Fiber cost



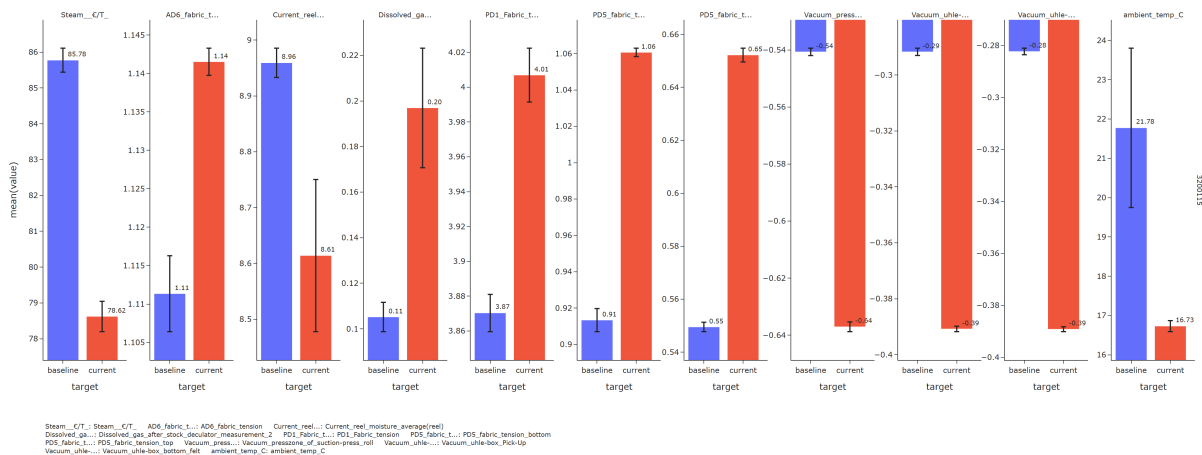
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Current reel moisture average(reel) decreased (-0.32)



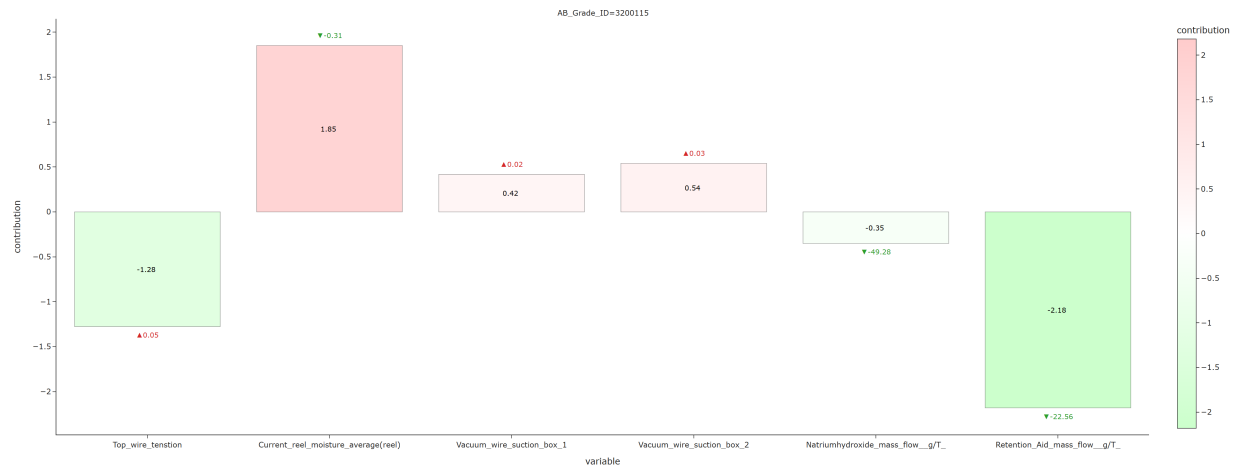
3.1.2 Steam cost



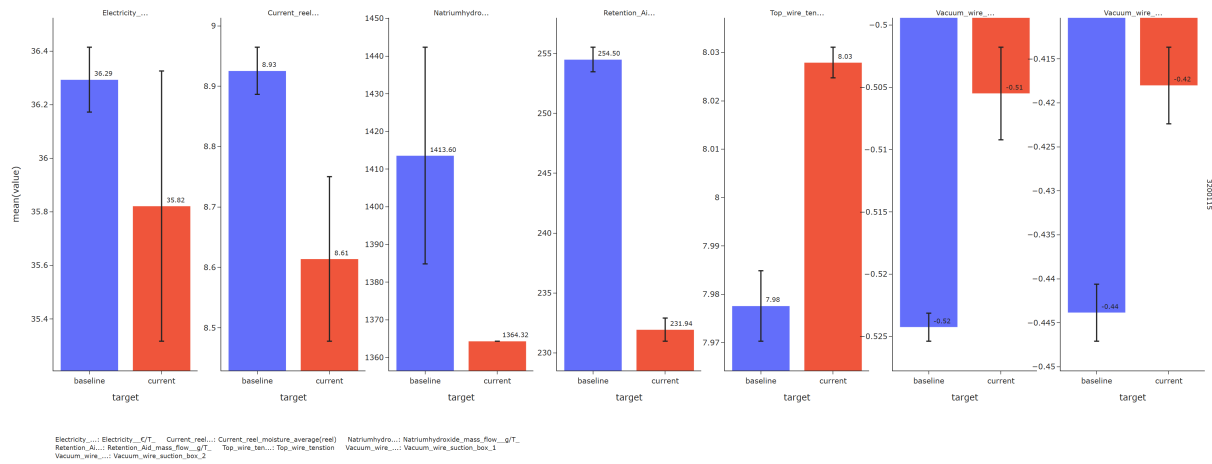
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - PD5 fabric tension top increased (+0.10) - Vacuum uhle-box bottom felt decreased (-0.11) - Vacuum uhle-box Pick-Up decreased (-0.10) - Dissolved gas after stock decudator measurement 2 increased (+0.09) **Cost-increasing drivers:** - Ambient temp C decreased (-5.05) - PD5 fabric tension bottom increased (+0.15)



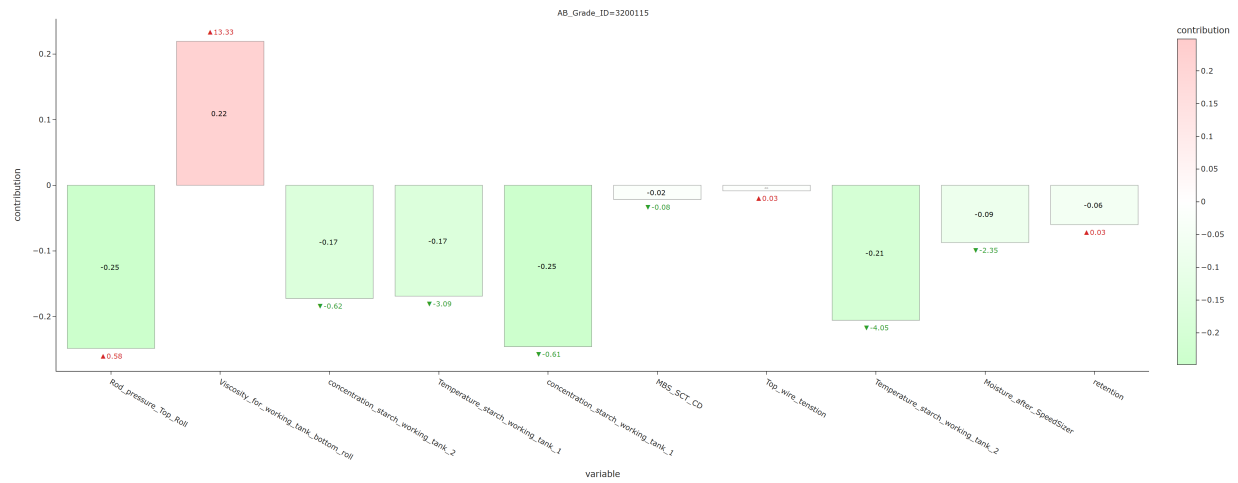
3.1.3 Electricity cost



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Retention Aid mass flow g/T decreased (-22.56) **Cost-increasing drivers:** - Current reel moisture average(reel) decreased (-0.31)



3.1.4 Starch cost



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Rod pressure Top Roll increased (+0.58) - Concentration starch working tank 1 decreased (-0.61)

